

Overfitting and Underfitting Analysis for Deep Learning Based End-to-end Communication Systems

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Abstract—In this paper, we study the deep learning (DL) based end-to-end transmission systems, then we present the analysis for the underfitting and overfitting phenomena which happen during the training of the neural networks (NNs). Different from the DL based image processing, the transmission systems require the bit error rate to be as low as 10^{-4} to achieve high reliability, thus the training errors induced by the underfitting and overfitting may greatly degrade the transmission reliability performances of DL-based communications. In considered DL-based communication systems, we propose to use the average, variance and minimum of transmitted signals' minimum Euclidean distance to estimate the effects of underfitting and overfitting on the error rate performances in terms of the energy per bit to noise power spectral ratio E_b/N_0 of signals. Furthermore, we propose to apply the regularization scheme to alleviate the overfitting issue. Simulations are performed to demonstrate the underfitting and overfitting analysis for transmission systems over the additive white Gaussian noise (AWGN) channel, and validate the improved performances by applying the regularization.

Index Terms—Deep learning; energy-to-noise ratio; end-to-end transmission; neural network training.

I. INTRODUCTION

Deep learning (DL) is a powerful tool which can learn complex linear or nonlinear mappings through training neural networks (NNs). In recent years, DL has received significant research interests in the communication field as a method of realizing the functions completed by conventional communication modules, such as the modulation recognition [1], the channel estimation [2], the decoding [3] etc. Moreover, DL has also been used to achieve intelligent communications [4].

To be more specific, [1] presents a DL approach to address the modulation recognition issue to improve the flexibility and accuracy by applying convolutional NNs to directly learn features from data instead of using expert feature based methods. In [2], NNs are applied in the channel estimation for beam space millimeter-wave massive multiple input and multiple output systems, which can achieve better performances with a small number of radio-frequency chains at the receiver and thus reduce the cost of hardware and power consumption. [3] proposes to use deep NNs to modify the belief propagation decoding algorithm in high-bandwidth digital content protection codes for higher bit error rate performances. Additionally, [4] presents a DL based transmission system that

can automatically find the optimal transmission schemes by jointly optimize the transmitter and receiver, in order to avoid the sub-optimal performances induced by the modular design. With the aid of DL, the communication systems can achieve higher performances and flexibility with intelligence, which also presents a new way of building optimal transmission systems.

When applying DL to build a transmission system, the whole communication system is considered as a blackbox and the transmitter as well as receiver components are jointly optimized by NN. In this way, the NN can be trained to automatically perform the optimal encoding, modulation, demodulation and decoding processing.

The basis of DL is the neural network. The training performance of NNs is vital to the performance of NN-aided applications since NNs need to learn and extract the features of the specific functions or mappings we want. Underfitting and overfitting are two major issues existent in the training stage [5]. The former underfitting problem means that the NN training is insufficient and the precision of the learning is low, while the overfitting problem means the learned mappings only perform well for training data and can not work well for the test data, thus the flexibility and adaptability are weak.

Underfitting and overfitting phenomena have been widely investigated in the machine learning, which has been presented and analyzed by researchers [5], [6]. Up to now, mainstream view considers that underfitting and overfitting are related to the complexity of NN architectures, the number and type of sample data used in training sets and the number of training epochs, which we will introduce in the next section.

In order to address the issues of the underfitting and the overfitting, several methods have been proposed. For example, [7] presents to increase the expressiveness of the latent distributions and use a higher capacity NN to solve underfitting in video prediction task. [8] applies dropout to improve the adaptabilities and robustness of NNs by probabilistically removing neurons from NN layers. And in [9], data augmentation is applied to reduce overfitting and improve accuracy by increasing the amount of training data with the information only in the training set. In addition, recent researches reveal that unlike classical view that complex NN architectures are

likely to cause overfitting, practices demonstrate that the NNs can still have excellent generalization performances if their complexities are higher than a threshold [10], which may promote the development of the machine learning theory and change the standard of NN model design.

However, as far as we know, there are no researches analyzing the underfitting and the overfitting training problems for communication systems, which is adverse to modify the training methods and improve the performances of DL based communication technologies.

In this paper, we analyze the underfitting and overfitting problems existent in the NN training, and propose to calculate the statistical characteristics of the minimum Euclidean distance of transmitted signals to estimate the influence of the underfitting and overfitting on the symbol error rate (SER) performances. Then, the simulations are provided to demonstrate the reliability performances versus the energy per bit to noise power spectral density ratios (E_b/N_0).

The main contributions are as follows: (1) we present the analysis of underfitting as well as overfitting problems for the DL based transmission systems; (2) we propose to calculate the statistical characteristics of transmitted signals' minimum Euclidean distance to estimate the influence of underfitting and overfitting on the SER performances; (3) we propose to apply the regularization method to improve the performances; (4) we provide simulations to demonstrate the reliability performances in terms of E_b/N_0 when underfitting and overfitting occur during the NN training.

The rest of the paper is organized as follows. Section II describes the basic principles of DL based transmission systems and the structure of the NN. In Section III, we give the definitions of underfitting and overfitting and introduce the analysis of them for the considered systems. Subsequently, Section IV presents the simulation results of NNs trained for different E_b/N_0 over additive white Gaussian noise (AWGN) channels to explore the influences of the training E_b/N_0 on the SER performances and demonstrate the underfitting and overfitting problems in the systems. Finally Section V concludes this paper.

II. DL-BASED END-TO-END TRANSMISSION SYSTEM

In this section, we describe the DL based end-to-end transmission systems and the NN structure used in these systems.

A. System Model

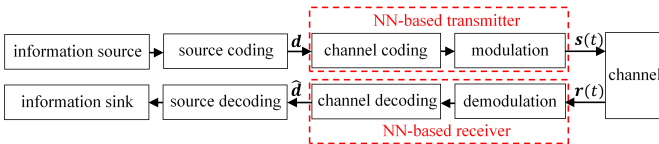


Fig. 1. The diagram of DL-based transmission systems.

Fig. 1 provides the diagram of the end-to-end DL-based transmission systems which transmit the information based on the NN aided transceivers. The presented DL-based transceiver

structure replaces the traditional transmission modules including the channel encoding and decoding, the modulation and demodulation modules.

Specifically, let $\mathbf{d}$ denote the message from end users and $\hat{\mathbf{d}}$ represent the estimates obtained from the receiver. In this DL-based information transmission system, the transmissions between $\mathbf{d}$ and $\hat{\mathbf{d}}$ seen as a blackbox in one NN. With the aim to achieve better transmission performances, the NN-aided systems are trained with the given samples to attain the mapping pattern embedded into the information transmissions. Using the resultant optimized parameters, the DL-based transmission system is used for optimal information transmissions.

Different from the conventional communication system, the considered DL-based transmission system consists of the transmitter component, the channel component and the receiver component. In the conventional communication system, wherein the received signals are obtained from the convolutional product of the channel response $\mathbf{h}(t)$ and the transmitted signals $\mathbf{s}(t)$, i.e.,

$$\mathbf{r}(t) = \mathbf{h}(t) * \mathbf{s}(t) + \mathbf{n}(t) \quad (1)$$

where $\mathbf{n}(t)$ represents the additive white Gaussian noise, and $*$ means convolution operation.

By contrast, in DL-based transceivers, there are no such explicit expressions to represent the convolutional product operations of the channel response and the transmitted signal. At the transmitter component, the message $\mathbf{d}$ to be sent is mapped into n numbers after several linear and nonlinear operations to make n discrete uses of the channel. Therefore, the transmitter component acts as a mapping $F(\cdot)$ and outputs the transmitted signal $\mathbf{s}(t) = F(\mathbf{d})$.

Subsequently, a specific channel environment is constructed as the channel component. When passing through the channel component, the transmitted signal $\mathbf{s}(t)$ undergoes the additive noises and multiplicative interferences, and the received signal $\mathbf{r}(t)$ is obtained as

$$\mathbf{r}(t) = I(\mathbf{s}(t)) \quad (2)$$

where $I(\cdot)$ is a function used to describe the channel component to simulate the additive noises of $\mathbf{n}(t)$ and channel responses of $\mathbf{h}(t)$.

Finally, in the receiver component, the estimates $\hat{\mathbf{d}}$ of the transmitted message $\mathbf{d}$ can be recovered from $\mathbf{r}(t)$ after the linear and nonlinear operations. Let $G(\cdot)$ represent the mapping from $\mathbf{r}(t)$ to $\hat{\mathbf{d}}$, namely, we have

$$\hat{\mathbf{d}} = G(\mathbf{r}(t)). \quad (3)$$

B. NN-aided DL-based Transmission System

To elaborate a bit further, Fig. 2 gives the NN architecture for DL-based transmission systems.

As shown in Fig. 2, when a message $\mathbf{d}$ is input to the transmitter component, it is firstly encoded as an one-hot vector $\mathbf{l}_d$. Then two fully connected (FC) layers with rectified linear unit (ReLU) and linear activation functions perform nonlinear and linear operations on $\mathbf{l}_d$ to realize encoding and

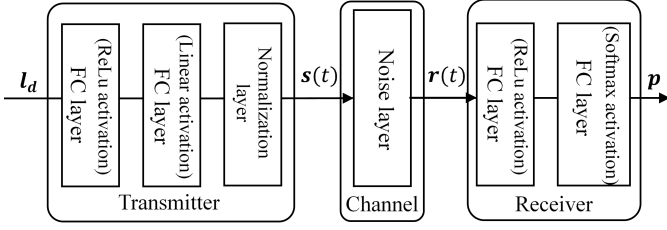


Fig. 2. The NN architecture used in DL-based transmission systems.

modulation processing, which can be expressed as

$$\mathbf{a}_0 = \sigma_0(W_0 \mathbf{l}_d + b_0) \quad (4)$$

$$\mathbf{x}(t) = \sigma_1(W_1 \mathbf{a} + b_1) \quad (5)$$

where W_0 , b_0 and W_1 , b_1 are trainable parameters used in the first and second FC layers, and σ_0 , σ_1 represent ReLU and linear activation functions respectively. $\mathbf{a}$ and $\mathbf{x}(t)$ are the outputs of the first and second FC layers. Before delivering the transmitted signal, the normalization layer normalizes $\mathbf{x}(t)$ to constrain the signal's energy within the fixed value n , namely, we have

$$\mathbf{s}(t) = \frac{\mathbf{x}(t)}{\|\mathbf{x}(t)\|_2} \times \sqrt{n} \quad (6)$$

The channel component follows the transmitter component, wherein the interferences and the noises are considered to construct the equivalent channel model.

At the receiver component, the received signals $\mathbf{r}(t)$ are processed by two FC layers with ReLU and softmax activation functions as follows

$$\mathbf{a}_2 = \sigma_0(W_2 \mathbf{r}(t) + b_2) \quad (7)$$

$$\mathbf{p} = \sigma_2(W_3 \mathbf{a}_2 + b_3) \quad (8)$$

where W_2 , b_2 and W_3 , b_3 are trainable parameters in the third and fourth FC layers, σ_2 is softmax activation function, $\mathbf{a}_2$ and $\mathbf{p}$ are the output of the third and fourth FC layers. $\mathbf{p}$ is a probability vector which represents the probabilities over all possible messages.

At last, the estimated message $\hat{\mathbf{d}}$ can be obtained by identifying the index of the largest element of $\mathbf{p}$.

C. NN Training for Transceivers

In order to achieve reliable and efficient information transmissions for DL-based systems, we need to train the NN to minimize the difference between the transmitted message $\mathbf{d}$ and the estimated message $\hat{\mathbf{d}}$.

When training NNs, we usually need three data sets: training sets, validation sets and test sets. Each kind of data sets contain different input data and corresponding expected output. Training sets are used to train the NNs to adjust their parameters and to learn the objective mappings. Validation sets, which have different but similarly distributed data, are used to test the accuracy rate of NNs for further adjustment of the training processing and hyperparameters. Finally, test sets are used to verify the effectiveness of NNs.

More explicitly, the difference between $\mathbf{d}$ and $\hat{\mathbf{d}}$ can be calculated with the loss function L to evaluate the loss values. Thus we could adjust trainable parameters through the gradient descent method for the loss function to reduce the difference between $\mathbf{d}$ and $\hat{\mathbf{d}}$. A widely used optimization algorithm is stochastic gradient descent (SGD), which can update the trainable parameters as follows

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} L(\mathbf{d}, \mathbf{p}) \quad (9)$$

where $\boldsymbol{\theta}$ is the trainable parameter set, and η is learning rate which can control the speed of gradient descent. By iteratively updating the inputs $\mathbf{d}$, calculating $\mathbf{p}$ and performing gradient descent operations, we can continuously train the NN, and $\hat{\mathbf{d}}$, which is obtained from $\mathbf{p}$, can gradually approach $\mathbf{d}$.

As a result, NN parameters can be determined by the training, thereby with the aid of the trained parameters, the DL-based systems can utilize the mapping to realize the channel encoding/decoding and modulation/demodulation functions.

III. ANALYSIS OF UNDERFITTING AND OVERFITTING PROBLEMS

In this section, we present the analysis of underfitting and overfitting problems for NN-aided DL based transmission systems.

A. Definitions of Underfitting and Overfitting Problems

Definition 1: Underfitting occurs when the model is not able to obtain a sufficiently low error value on the training set.

Definition 2: Overfitting occurs when the gap between the training error and test error is too large.

As stated in Definition 1 and Definition 2 [6], underfitting occurs when the NNs can not capture the characteristics of training sets and can not fit the target mappings well, which will lead to low accuracy rates and high loss values for all three data sets. The possible factors that induce the underfitting problem are that NN architectures are too simple and have too few hidden layers or trainable parameters, which makes the NNs not powerful enough to capture complex characteristics of data [5]. Another possibility is that the NNs just have not been trained enough to learn the characteristics or distributions of data.

On the other hand, overfitting occurs when the NNs are overly dependent on training sets and learn wrong mappings which work well in training sets but perform poorly in validation or test sets. There are several possible reasons for this problem. If the NN architectures are too complicated in a certain range while the number of training data is inadequate, NNs may learn the characteristics of training data too well that they focus on specific features of training sets too much and ignore the common features [5] [8]. If the characteristic distributions of training sets are different from that of validation or test sets, the NNs are also unable to learn adaptable mappings. Both situations can lead to the overfitting problem.

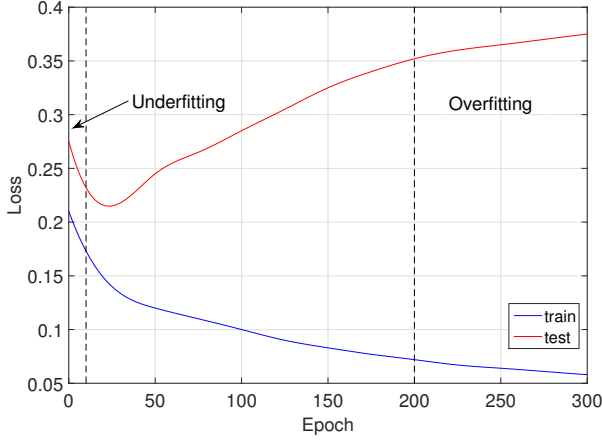


Fig. 3. An example of underfitting and overfitting problems.

B. Problem Analysis

Fig. 3 illustrates an example of the problems of the underfitting and the overfitting. In this example, we try to train a long short-term memory (LSTM) NN to learn the pattern of the dynamic watermarking. To be more specific, we set the bits from an MP3 file as input data and their watermarked signals as the expected outputs of the training set, while the test set uses bits from another MP3 file as the inputs and their watermarked signals as the expected outputs. Thus overfitting possibly occurs because the two data sets have different characteristic distributions. When training the NN, the Adam algorithm [11] with the recommended default learning rate 0.001 is chosen to be the optimization algorithm, and the mean squared error (MSE) is applied as loss function.

It can be seen from the figure that at the beginning of the training processing, the loss values of the NN in training set and test set are large, which demonstrates that the underfitting occurs in the first few epochs since it has not been trained enough to learn the mapping. Then, as the training proceeds, the loss values decrease in both sets for the NN has learnt more knowledge from the training data. Finally, because the training set we provide for the NN has different characteristic distributions from those of test set in this example, the loss values continuously decrease in the training set while increasing in the test set. In this case, the overfitting occurs since the system can achieve satisfactory reliability performances in the training set but does not work well in the test set.

In the DL-based transmission systems, the training set which consists of all possible messages will not induce the problem of the overfitting or the underfitting, since the three kinds of data set have accommodated all possible messages. However, the additive noise and the multiplicative interference may have dramatic variations in real wireless systems, which bring a big challenge for the NN training.

Next, we propose to calculate the statistical characteristics, including the minimum, the average and the variance, of the minimum Euclidean distance of transmitted signals to evaluate the underfitting and the overfitting degree for DL-

based transmission systems.

C. Performance Evaluation Based on Minimum Euclidean Distance

In the considered DL-based transmission system, the numbers of transmitted bits per symbol, the number of channels and the energy of transmitted signals remains the same, thus the encoded and modulated transmitted signals $s(t)$ can be treated as a kind of spherical codes, since every symbol of the codes has the same norm.

Considering that the spherical code is optimal when all codes with the same cardinality and dimensionality have the maximal minimum normalized Euclidean distance [12] [13], we propose to use the statistical characteristics of transmitted signals' minimum Euclidean distance to estimate the performances and the degrees of the underfitting and the overfitting problems. In order that all transmitted signals have the maximal minimum Euclidean distance, the minimum and average of their minimum Euclidean distance are expected to be high while the variance is expected to be low to keep a stable performances.

TABLE I
STATISTICAL CHARACTERISTICS OF TRANSMITTED SIGNALS' MINIMUM EUCLIDEAN DISTANCE AFTER TRAINING WITH DIFFERENT E_b/N_0 OVER AWGN CHANNELS

training E_b/N_0	minimums	averages	variances
11dB	2.672	2.910	0.0221
7dB	2.904	3.140	0.0208
3dB	3.268	3.393	0.0132
-1dB	3.047	3.287	0.0241
-5dB	2.933	3.282	0.0368
-9dB	2.585	2.867	0.0379

Table I evaluates the minimums, averages and variances of the minimum Euclidean distances of transmitted signals $s(t)$ which have been recorded. Moreover, as shown in the table, when the training E_b/N_0 decreases from 11dB to 3dB, the minimum and the average increase and the variance decreases. This observation is in accordance with that shown in Fig. 4 to be presented in the next section.

D. Regularization to mitigate overfitting

In order to mitigate the overfitting or underfitting effects for DL based systems, we propose to apply the regularization method [14] to alleviate the overfitting, which typically modifies the loss function of NN by adding a penalty term on the complexity of the model. Here we use the L2-norm regularization to combating the overfitting.

To be more explicit, we propose to modify the loss function as

$$L^*(\mathbf{d}, \mathbf{p}, \boldsymbol{\theta}) = L(\mathbf{d}, \mathbf{p}) + \lambda \sum_i \theta_i^2 \quad (10)$$

where $L^*(\cdot)$ is the modified loss function, $L(\cdot)$ is the original loss function, θ_i means the i th element of $\boldsymbol{\theta}$ and λ is a hyperparameter to adjust the strength of regularization.

From the equation we can see that the L2-norm regularization restricts the complexity of NN by shrinking the value

of trainable parameters. With the modified loss function, the NN are forced to learn to implement the expected mapping. Thus the NN is prevented from using complicated function to fit outliers in the training set, and the effects of the overfitting can be alleviated. However, when underfitting occurs, it means the NN is unable to learn the features of the training set, thus restricting the complexity of NN is useless to improve the performances in that situation, and regularization can not work for underfitting systems.

In the next section, we will provide simulation results to demonstrate the effects of the underfitting and the overfitting as well as the regularization on the error rate performances.

IV. SIMULATIONS AND ANALYSIS

In this section, simulations are performed to investigate the symbol error rate performances when underfitting and overfitting occur. In the following simulations, we assume that 16 possible messages (4 bits) are transmitted to the receiver over 7 channels. The NN architecture used in the DL based transmission systems is the same as the one in [4] as given in Fig. 2, where the signals' energy is constrained to be a fixed value $\|s(t)\|_2^2 = 7$.

The NNs are trained in different E_b/N_0 , the Adam algorithm with the default learning rate 0.001 is selected as the optimization algorithm, and the categorical crossentropy is applied as the loss function. Each NN is trained 5000 epochs. Since the NNs used in the simulations are not complex, a central processing unit (CPU) is enough for the training tasks. In addition, intel Core i5-7300HQ is used in our simulations.

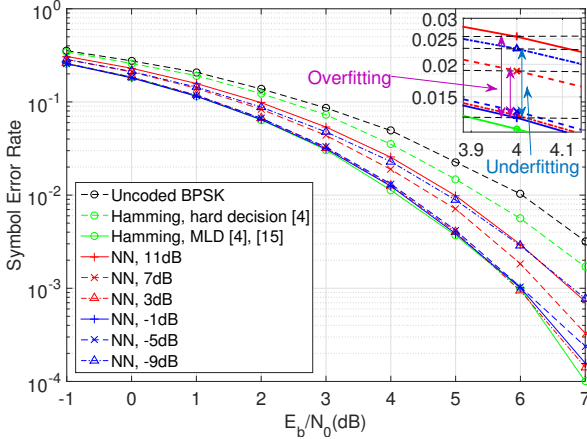


Fig. 4. Comparison of symbol error rate between NNs trained in different E_b/N_0 over AWGN channels.

Fig. 4 compares the symbol error rate performances of NNs which are trained in different E_b/N_0 over AWGN channels. We also plot the performances of conventional communication systems using the binary phase shift keying (BPSK) modulation and Hamming encoding method with either binary hard-decision decoding or maximum likelihood decoding (MLD) [15] as benchmarks for comparisons. The benchmarks are used in [4] as well. It can be seen that training NNs with different E_b/N_0 will lead to different reliability performances of the

DL-based transmission systems. In the inset, the blue double arrows indicate the performance degradations resulted from the underfitting, and the purple ones indicate the performance penalties caused by the overfitting when the testing E_b/N_0 is 4dB.

For example, when the training E_b/N_0 is as high as 11dB, the NN only learns as a simple autoencoder and only little knowledge can be extracted about how to suppress AWGN. In this case, the overfitting problem occurs. In addition, when the training E_b/N_0 has the moderate value, such as 3dB, the NN achieves the best performances since it learns a maximum knowledge about how to suppress the noise within its architecture capacity, and the performances are similar to that of systems using the BPSK modulation and Hamming encoding method with the MLD. Moreover, when the training E_b/N_0 is as low as -5dB, the DL-based communication system still works well with lower E_b/N_0 value, while the reliability performance degrades with higher E_b/N_0 value. The possible reason is that too much training data are severely interfered, thus the NN may focus on extreme cases instead of learning overall distributions. In this case, the overfitting problem occurs again.

Furthermore, when the training E_b/N_0 falls to -9dB, the interference is too strong that the NN can not learn much about the noise characteristics, which induces serious performance degradations, which can be interpreted as a kind of underfitting problem.

Therefore, when an appropriate value of E_b/N_0 is used for training, NNs can learn the encoding and modulation better and prevent the overfitting for the training E_b/N_0 of large values as well as the underfitting for smaller values of training E_b/N_0 .

To elaborate a bit further, recall Table I, we can notice that when the training E_b/N_0 decreases from 3dB to -5dB, the minimum and average begin to decrease and the variance starts to increase. However, the reduction of the average is relatively small, which may explain why the symbol error rate performances of NNs trained in 3 to -5dB are similar to those trained with lower E_b/N_0 value. In addition, since the growth in variance is relatively obvious, it is not strange that NNs trained in lower E_b/N_0 value perform worse than those trained with higher E_b/N_0 value. Finally, when the training E_b/N_0 decreases to -9dB, the noise is too severer that NNs can not learn a proper encoding and modulation scheme and the minimum as well as average drop rapidly. This observation matches that obtained in Fig. 4 since the underfitting occurs when trained with lower E_b/N_0 value.

At last, we apply L2-norm regularization in the NN to improve the symbol error rate performances. As shown in Fig. 5, after applying L2-norm regularization to modify the loss function, the NN can achieve better performances when training in 11dB, which demonstrates that regularization can help combat the overfitting in DL based transmission systems. However, applying L2-norm regularization can not improve the performances when the NN is trained in 3dB or -9dB. The results illustrates that regularization can alleviate the

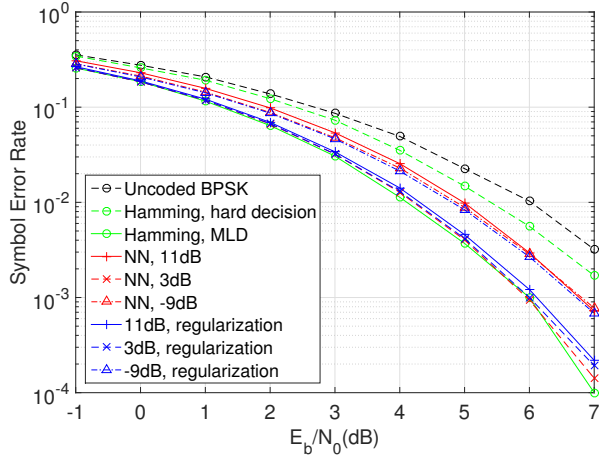


Fig. 5. Comparison of symbol error rate between NNs trained with and without L2-norm regularization over AWGN channel.

overfitting but does not work for underfitting systems. Moreover, it has negative effect that it might degrade the optimal performances of NN.

V. CONCLUSION

In this paper, we analyze the underfitting and the overfitting problems occur in DL-based transmission systems. Based on the DL-aided end-to-end communication system model, we present the NN architecture used to complete the channel encoding/decoding and modulation/demodulation functions. Subsequently, we present the underfitting and overfitting phenomena appearing when training NNs, and propose to use the statistical characteristics of the minimum Euclidean distance of transmitted signals to measure the underfitting and the overfitting degree. Then, simulations are performed to investigate the symbol error rate performances in terms of E_b/N_0 . The results demonstrate that the observation based on the analysis of the minimum, the average and the variance is in accordance with the simulations. Besides, we also attempt to apply L2-norm regularization in NN to mitigate the overfitting influences on the error rate performances. Last but not the least, the presented analysis of the underfitting and overfitting mainly utilizes the parameter employed at the transmitter component,

further researches can be extended to more parameters of the channel component and the receiver component.

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REFERENCES

- [1] T. J. O'shea, J. Corgan, and T. C. Clancy, "Convolutional radio modulation recognition networks," in *Proc. International conference on engineering applications of neural networks*, 2016, pp. 213–226.
- [2] H. He, C.-K. Wen, S. Jin, and G. Y. Li, "Deep learning-based channel estimation for beamspace mmwave massive mimo systems," *IEEE Wireless Commun. Lett.*, vol. 7, no. 5, pp. 852–855, 2018.
- [3] E. Nachmani, Y. Be'ery, and D. Burshtein, "Learning to decode linear codes using deep learning," in *Proc. 54th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, 2016, pp. 341–346.
- [4] T. O'Shea and J. Hoydis, "An introduction to deep learning for the physical layer," *IEEE Trans. Cogn. Commun. Netw.*, vol. 3, no. 4, pp. 563–575, 2017.
- [5] L. N. Smith, "A disciplined approach to neural network hyperparameters: Part 1—learning rate, batch size, momentum, and weight decay," *arXiv preprint arXiv:1803.09820*, 2018.
- [6] I. Goodfellow, Y. Bengio, and A. Courville, *Deep learning*. MIT press, 2016.
- [7] L. Castrejon, N. Ballas, and A. Courville, "Improved conditional vmns for video prediction," *arXiv preprint arXiv:1904.12165*, 2019.
- [8] G. E. Hinton, N. Srivastava, A. Krizhevsky, I. Sutskever, and R. R. Salakhutdinov, "Improving neural networks by preventing co-adaptation of feature detectors," *arXiv preprint arXiv:1207.0580*, 2012.
- [9] L. Perez and J. Wang, "The effectiveness of data augmentation in image classification using deep learning," *arXiv preprint arXiv:1712.04621*, 2017.
- [10] M. Belkin, D. Hsu, S. Ma, and S. Mandal, "Reconciling modern machine learning and the bias-variance trade-off," *arXiv preprint arXiv:1812.11118*, 2018.
- [11] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- [12] A. E. Geyer, R. Nikjah, S. A. Vorobyov, and N. C. Beaulieu, "Euclidean and space-time block codes: Relationship, optimality, performance analysis revisited," *IEEE Trans. Commun.*, vol. 63, no. 8, pp. 2912–2923, 2015.
- [13] T. Ericson and V. Zinoviev, *Codes on Euclidean spheres*. North-Holland Publishing Co., Amsterdam: Elsevier, 2001, vol. 63.
- [14] Z. Luo, S. Cai, J. Gao, M. Zhang, K. Y. Ngiam, G. Chen, and W. Lee, "Adaptive lightweight regularization tool for complex analytics," in *Proc. IEEE 34th International Conference on Data Engineering (ICDE)*, April 2018, pp. 485–496.
- [15] S. Host, *Channel Coding*. IEEE, 2019, pp. 169–212. [Online]. Available: <https://ieeexplore.ieee.org/document/8698976>